
DV2626/DV2599 Machine Learning

Lecture 2: Foundations

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(some slides from Niklas Lavesson)

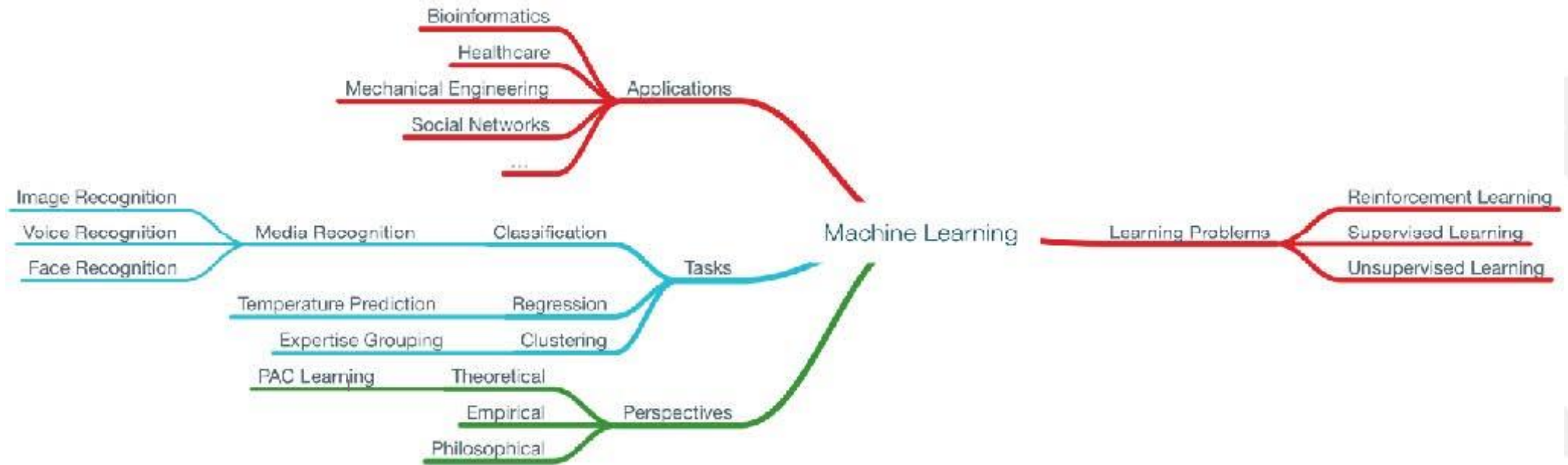
Outline

- ML Overview
- Terminology
- Binary classification
- Beyond binary classification
 - Regression, clustering
- Concept learning
- References

Growth of Machine Learning

- Machine learning is preferred approach to
 - Speech recognition, Natural language processing
 - Computer vision
 - Medical outcomes analysis
 - Robot control
 - Computational biology
 - ...
- This trend is accelerating
 - Improved machine learning algorithms
 - Improved data capture, networking, faster computers
 - Software too complex to write by hand
 - New sensors / IO devices
 - Demand for self-customization to user, environment
 - It turns out to be difficult to extract knowledge from human experts

Machine Learning Overview (1)



¹Probably Approximately Correct Learning

Machine Learning Overview (2)

- type of tasks
 - classification (e.g., face recognition)
 - regression (e.g., temperature prediction)
 - clustering (e.g., gene functional analysis)
- use of knowledge
 - empirical (knowledge-free)
 - analytical (knowledge-guided)
- representation
 - attribute-based (feature vector)
 - relational (first-order logic)
- type of learning (feedback)
 - supervised (labeled examples)
 - unsupervised (unlabeled examples)
 - reinforcement (reward)

Terminology (1)

- The objects of interest in machine learning are generally referred to as **instances**
- The set of all possible instances is called the **instance space**
- A **training set** contains a number of labelled instances (**examples**)
- A **test set** contains examples that are only used to test or evaluate the model (that is, they are not used for training)
- Instances are usually described by a fixed number of **features**
- The **domain** of a feature is the set of possible values
 - If x_i is the size of the domain of feature i and n is the number of features, then the size of the instance space is: $x_1 \times \dots \times x_n$

Terminology (2)

- A model is a mapping between the instance space and the **output space**
 - A **classifier** is a mapping between the instance space and the set of possible classes
 - A **regression function** is a mapping between the instance space and the set of real numbers
- Background 2.1 in the main literature contains a brief review of some useful concepts from discrete mathematics
- Table 2.1 in the main literature contains descriptions of some common predictive machine learning scenarios

Machine Learning Tasks

- Task refers to whatever it is machine learning is intended to improve performance of
- Different tasks require different types of models to be learned:
 - Classification tasks - Classifiers
 - Regression tasks - Regression functions
 - Clustering tasks - Clusterings or Cluster models
- Many models exist for each model type and which one to use depends on the context

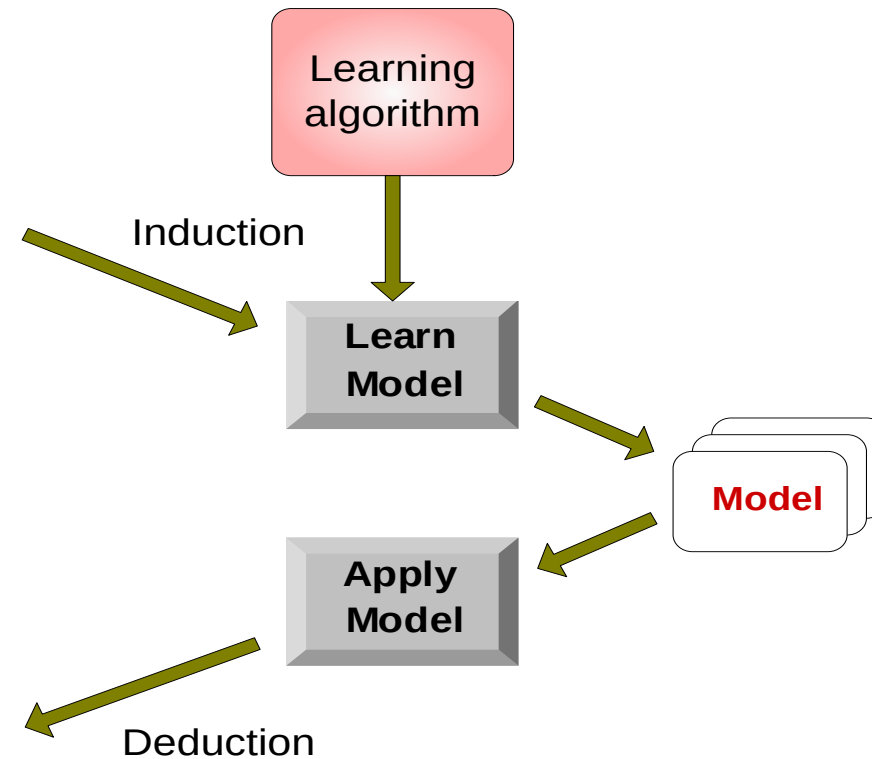
Illustration of classification task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Classification (1)

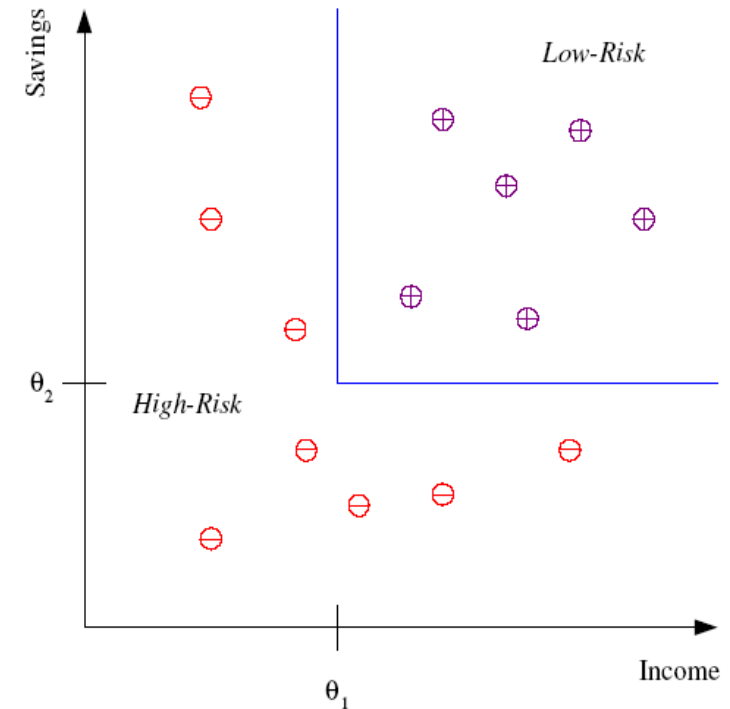
- **Classification** is the most common task in machine learning
- A **classifier** is a mapping between the instance space and a finite and usually small set of **class labels**
- Instances are generally assumed to be drawn from a well-defined but unknown probability distribution that governs the likelihood of occurrence of different instances in the instance space
 - We often assume that examples are drawn from the same distribution (and thus that the training and test sets are **representative samples** of the problem)

Classification (2)

- In the simplest case we have only two classes which are usually referred to as **positive** and **negative**.
- Two-class classification is often called **binary classification**
- Spam e-mail filtering is a good example of binary classification, in which spam is conventionally taken as the positive class, and ham as the negative class.
- Other examples of binary classification include medical diagnosis (the positive class here is having a particular disease) and credit scoring (the positive class may be to have a low-risk).

Binary Classification Example

- Example: Credit scoring
- Differentiating between **low-risk** and **high-risk** customers from their *income* and *savings*
- Model: IF *income* $> \theta_1$ AND *savings* $> \theta_2$
THEN **low-risk** ELSE **high-risk**

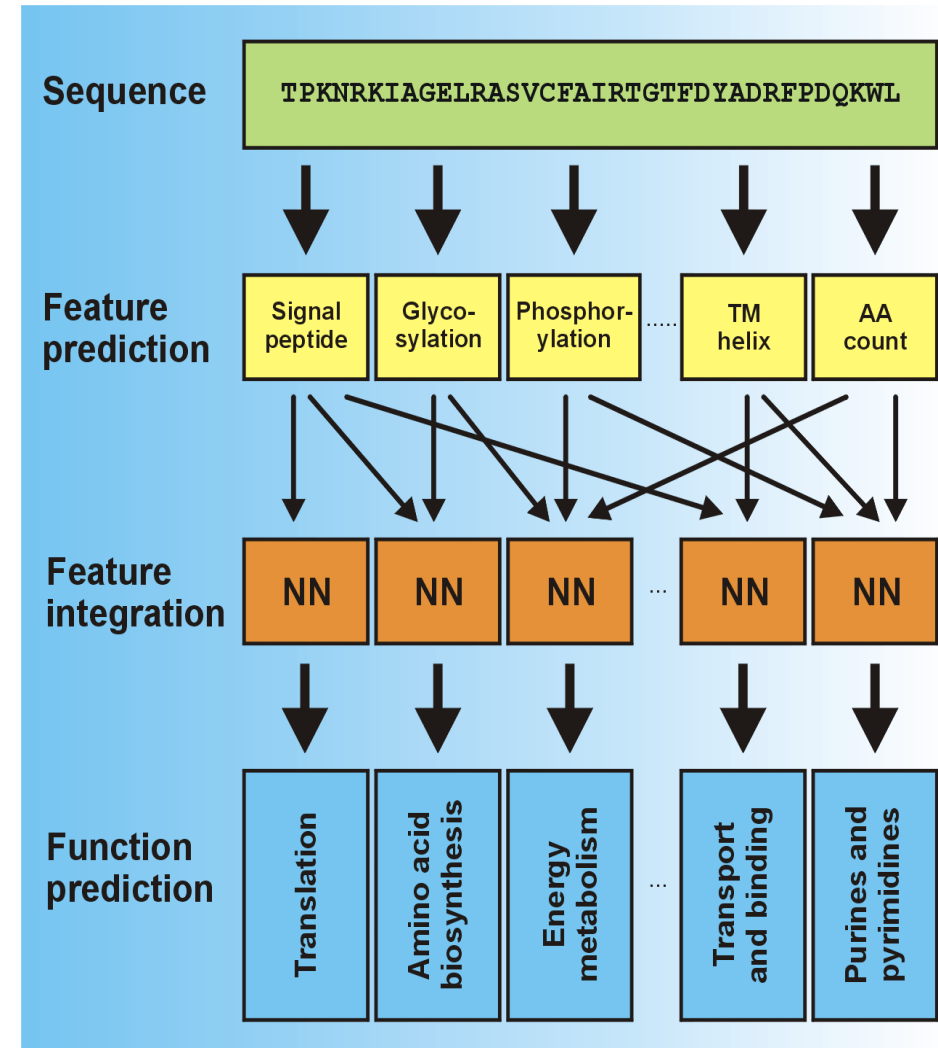


Scoring, ranking, and probability estimating classifiers

- Sections 2.2 and 2.3 introduce three additional, and sometimes quite useful, variants of classifiers:
 - The **scoring classifier** outputs a vector of scores instead of a class label. Each score is associated with a different class label and denotes the strength of belief that the label is correct
 - The **ranking classifier** orders instances based on strength of belief in that they belong to the positive class. A ranking error occurs when one instance receives a lower value than another even though the first instance is positive and the second is negative
 - A **probability estimation classifier** is a scoring classifier that outputs a probability vector over classes

Probability Estimation Classifier Example

- Example: Protein function prediction
- Predict as many biologically relevant features as we can from the sequence
- Train artificial neural network for each category
- Assign a probability for each category from the NN outputs



Evaluation of classifier performance (1)

- The performance of a classifier is computed by letting it classify all the instances of the test set
 - The **accuracy** is obtained by dividing the number of correctly classified instances by the total number of classified instances
 - Similarly, the **error rate** is obtained by dividing the number of incorrectly classified instances by the total number of classified instances
- Notice that the magnitude of the error is unimportant here; either a classification of an instance is correct or it isn't
 - In regression, the standard approach to calculate the prediction performance is to compute the magnitude of the error

Evaluation of classifier performance (2)

- It is convenient, and sometimes necessary, to distinguish performance on the classes:
 - **true positives** - TP - correctly classified as belonging to the positive class
 - **false positives** - FP - incorrectly classified as belonging to the positive class
 - **true negatives** - TN - correctly classified as belonging to the negative class
 - **false negatives** - FN - incorrectly classified as belonging to the negative class

Evaluation of classifier performance (3)

- True positive rate, $TPR = TP / (TP + FN)$
- **False positive rate** (false alarm rate), $FPR = FP / (FP + TN)$
- False and true negative rate (TNR) are analogously defined
- TPR and TNR are per-class accuracy measures and are usually referred to as **sensitivity** and **specificity**
- So, when is accuracy inappropriate? For example, when class imbalance is high and the positives are few but important
 - Imagine a spam detector classifying 99 out of 100 e-mails correctly as legitimate (negative) but misclassifying the one spam as legitimate as well (false negative). This classifier would still have an accuracy of 0.99

Contingency table (or confusion matrix)

	Predicted +	Predicted -	
Actual +	TP	FN	TP + FN
Actual -	FP	TN	FP + TN
	TP + FP	FN + TN	Total

$$\text{TPR (sensitivity)} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{FPR (false alarm rate)} = \text{FP} / (\text{FP} + \text{TN})$$

$$\text{TNR (specificity)} = \text{TN} / (\text{TN} + \text{FP}) = 1 - \text{false alarm rate}$$

Precision and F-measure

- The sensitivity (**hit rate** and **recall**) is a statistical measure of how well a binary classification test correctly identifies a condition.
- The specificity is a statistical measure of how well a binary classification test correctly identifies the negative cases
- **Precision** is probability that a positive prediction is correct,
 $precision = TP / (TP + FP)$
- The F-measure is the harmonic mean of precision and recall,
 $F\text{-measure} = (2 \times precision \times recall) / (precision + recall)$

Contingency table (or confusion matrix)

	Predicted +	Predicted -	
Actual +	TP	FN	TP + FN
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	TP + FP	FN + TN	Total

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$$\text{precision} = \text{TP} / (\text{TP} + \text{FP})$$

Contingency table (or confusion matrix)

	Predicted +	Predicted -	
Actual +	TP (Hit)	FN (Miss)	TP + FN
Actual -	FP (False alarm)	TN (Correct Rejection)	FP + TN
	TP + FP	FN + TN	Total

$$\text{TPR (sensitivity)} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{FPR (false alarm rate)} = \text{FP} / (\text{FP} + \text{TN})$$

$$\text{TNR (specificity)} = \text{TN} / (\text{TN} + \text{FP}) = 1 - \text{false alarm rate}$$

$$\text{precision} = \text{TP} / (\text{TP} + \text{FP})$$

Coverage plots

- Coverage plots are an excellent way to visualize the classification performance
 - The y-axis depicts positives and the x-axis depicts negatives
 - The range of y is from 0 to the number of positives
 - The range of x is from 0 to the number of negatives
 - We order instances based on the classifier's labelling; positives first and then negatives
 - For each true positive, we move up one step and for each false positive, we move right one step,
 - then we take the negatives and move right for each true negative and up for each false negative
 - We can now represent a classifier's performance with a specific point in the plot (see Figure 2.3 in the main literature)

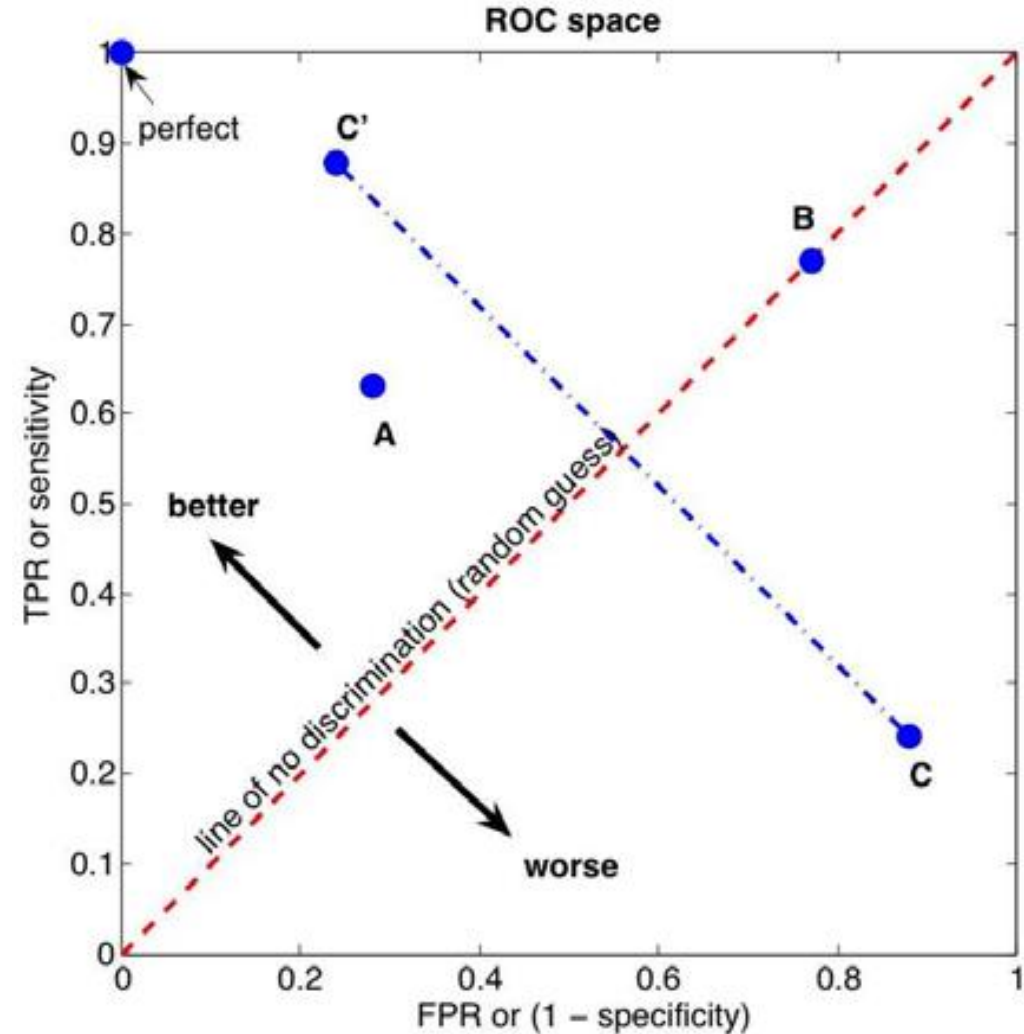
ROC curve (1)

- ROC¹ graphs have long been used in signal detection theory to depict the tradeoff between hit rates and false alarm rates over noisy channel
- Recent years have seen an increase in the use of ROC graphs in the machine learning community
- A useful technique for organizing classifiers and visualizing their performance
- Especially useful for domains with skewed class distribution and unequal classification error costs

¹Receiver Operating Characteristics

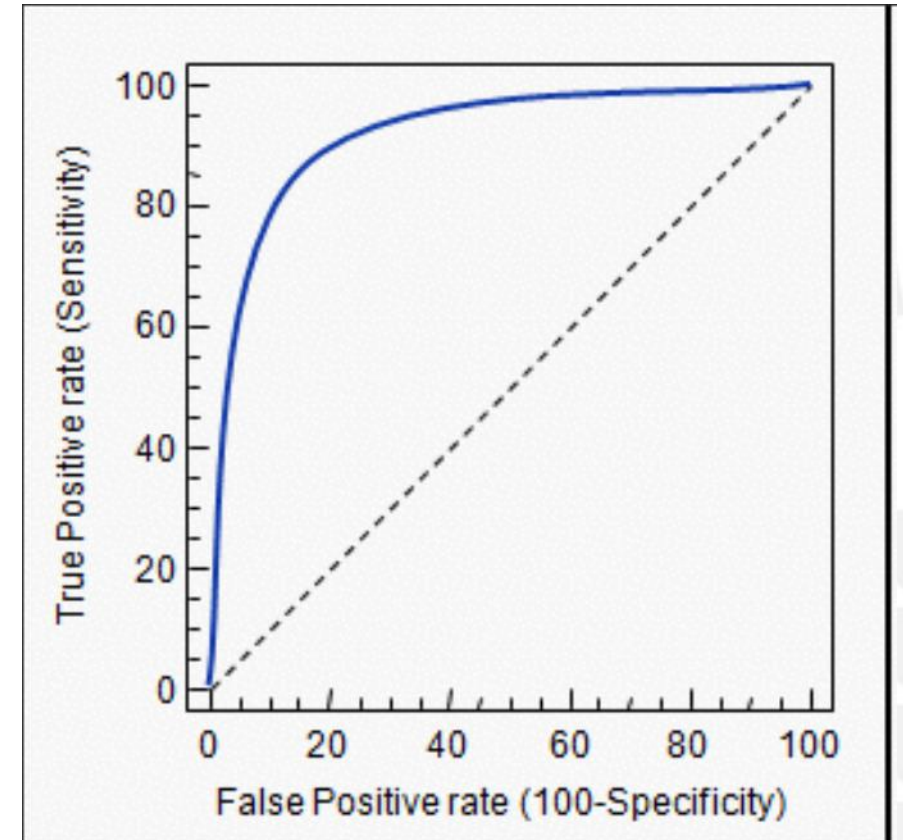
ROC space

ROC curve is a plot of TPR against FPR which depicts relative trade-offs between benefits (true positives) and costs (false positives)



ROC curve (2)

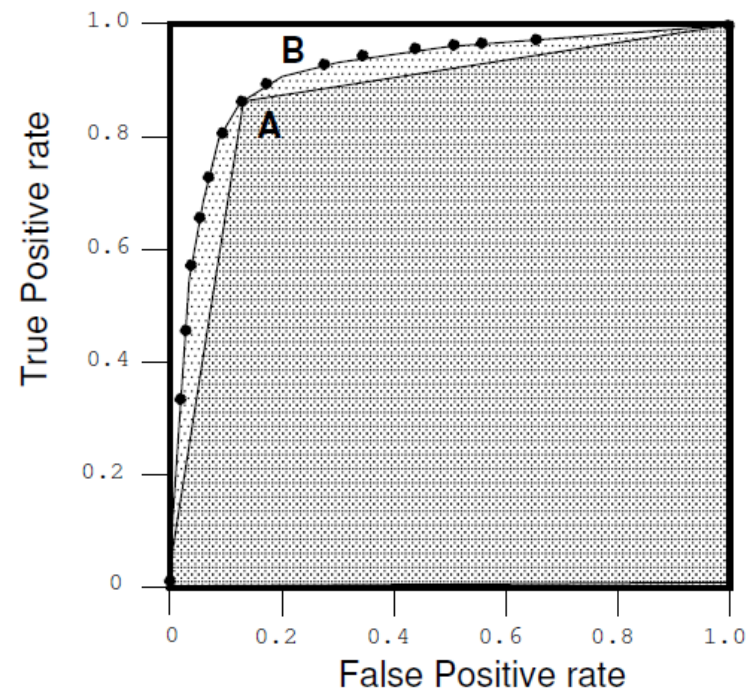
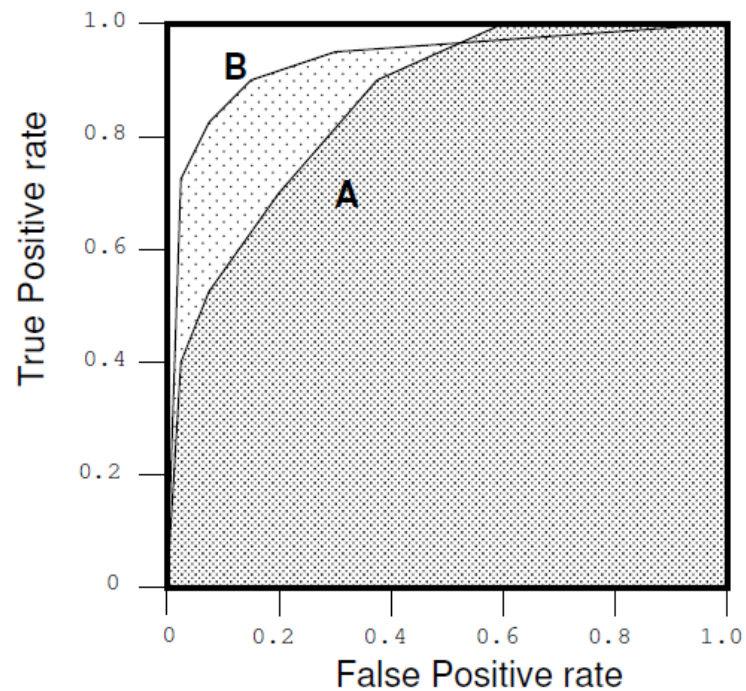
- A discrete classifier produces an (FPR, TPR) pair corresponding to a single point in ROC space
- A ranking or scoring classifier can be used with a threshold to produce a discrete classifier
- Plotting the ROC point for each possible threshold value results in a curve



ROC curve (3)

- ROC curves offer a more complete picture of the performance of the classifier
- We often do not know costs, or changing class distributions
- To compare classifiers we may want to reduce ROC performance to a single scalar value representing expected performance
- A common method is to calculate the area under the ROC curve, abbreviated AUC

ROC curve example



$AUC(B) > AUC(A)$: classifier *B* has better average performance than *A*

Regression

- A function estimator, or regressor, is a mapping from the instance space to the set of real numbers
- The regression learning problem is to learn a function estimator from examples
- To avoid overfitting, the number of parameters estimated from the data must be considerably less than the number of data points
- The most obvious loss function for evaluation is quadratic loss, which is used to compute the mean squared error (MSE) on unseen test data

Regression Task Example

- Example: Price of a used car

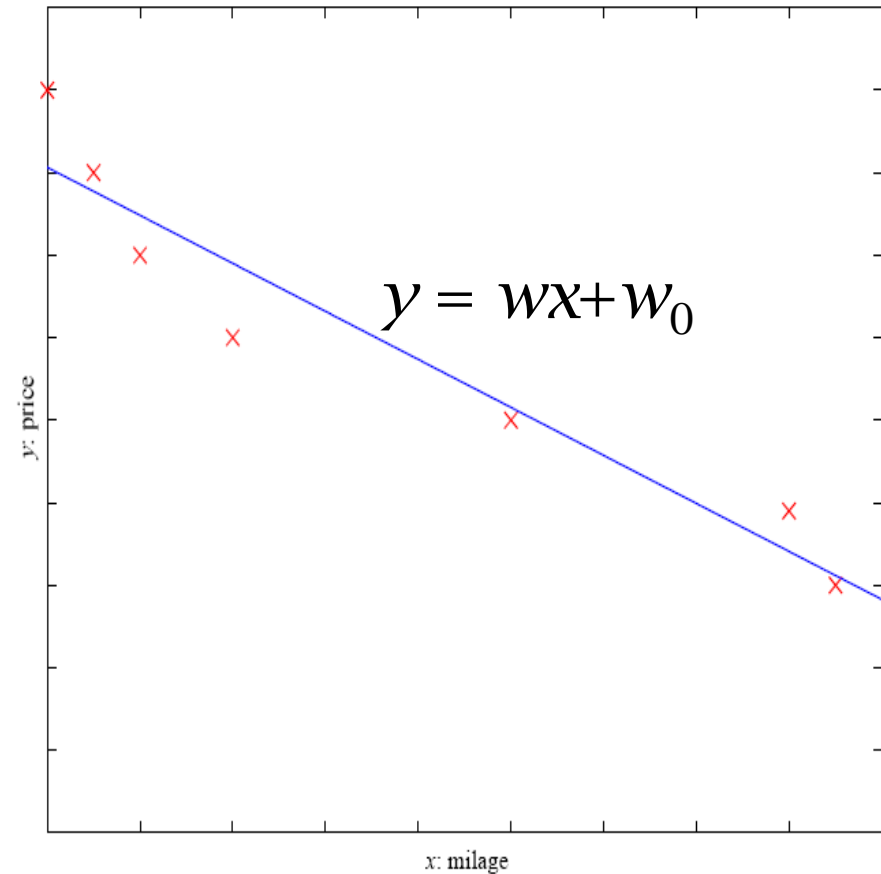
- x : car attributes

y : price

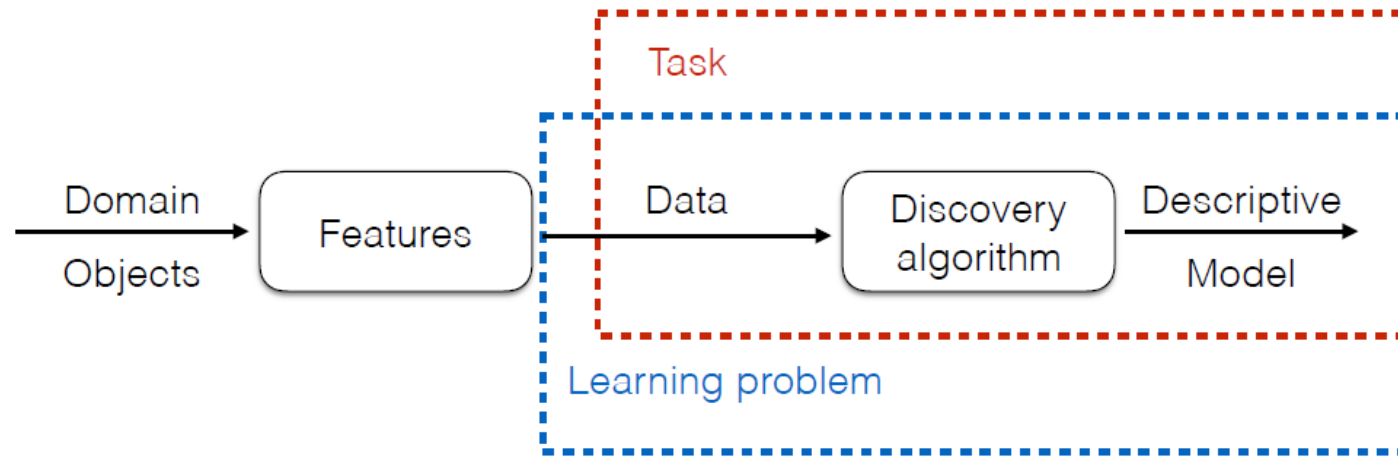
$$y = g(x \mid \vartheta)$$

$g(\cdot)$: model,

ϑ : parameters

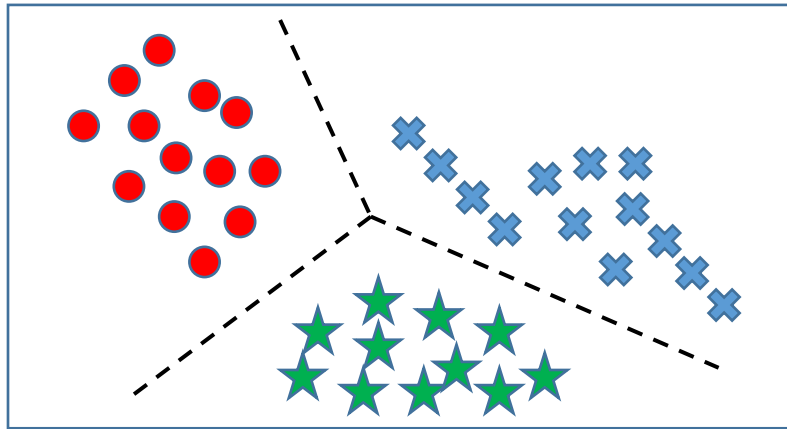


Unsupervised Learning

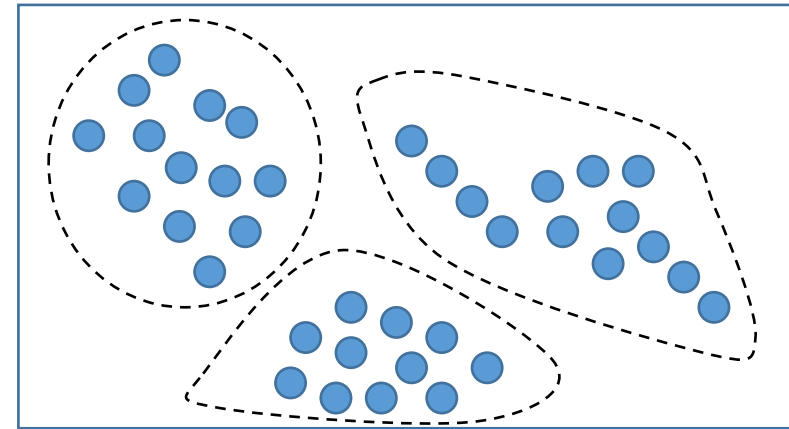


In descriptive learning the task and learning problem coincide: we do not have a separate training set, and the task is to produce a descriptive model from data - Flach (2012)

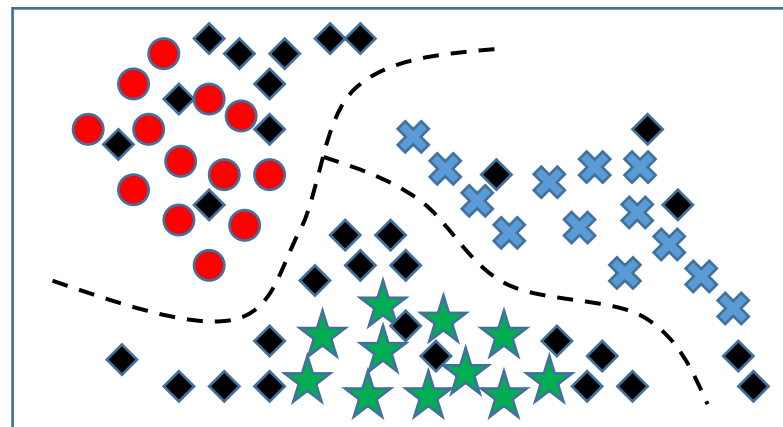
Supervised vs Unsupervised



Supervised learning



Unsupervised learning



Semi-supervised learning

Unsupervised Learning

Learn patterns from (unlabeled) data.

- Approaches
 - clustering (similarity-based)
 - density estimation (e.g., EM¹ algorithm)
- Performance Tasks
 - understanding and visualization
 - anomaly detection
 - information retrieval
 - data compression

¹Expectation–Maximization

Clustering

- Clustering is used to divide data into coherent groups (clusters)
 - Coherence here means that, on average, two instances from the same cluster have more in common, are more similar, than two instances from different clusters
 - Similarity (and consequently dissimilarity) can be measured in many different ways
 - *which way is appropriate is based on context*
 - *The Euclidean distance is the most obvious distance measure if the features are real-valued*
- The most common general approach is to minimise the within-cluster scatter, that is, to find clusters whose center points allow for the smallest sums of distances between the center points and remaining cluster instances

Concept Learning (1)

- Logical machine learning models originate from computer science and divide the instance space into segments with logical expressions
- The goal is to find a segmentation such that the data in each segment is more homogeneous, with respect to the task being solved
- Logical models generally fall into two categories: tree models and rule models
- Concept learning, due in large part to Bruner et al. (1956), concerns the learning of logical expressions, or concepts, from examples
 - Concept learning only learns descriptions of the positive class

Concept Learning (2)

- Task: *learning a category description (concept) from a set of positive and negative training examples.*
 - Concept may be an event, an object ...
- Target function: a *boolean function* $c: X \rightarrow \{0, 1\}$
- Experience: a *set of training instances* $D: \{\langle x, c(x) \rangle\}$
- A search problem for best fitting hypothesis in a hypotheses space
 - The space is determined by the choice of representation of the hypothesis (all boolean functions or a subset)

Concept Learning Example (1)

- Concept to be learned:
 - Days in which Mark can enjoy water sport
 - Attributes (features):
 - Sky*: Sunny, Rainy
 - Wind*: Strong, Weak
 - AirTemp*: Warm, Cold
 - Water*: Warm, Cool
 - Humidity*: Normal, High
 - Forecast*: Same, Change
- Instances in the training set (out of the 96 possible):

Sky	Temp	Humid	Wind	Water	Forecast	EnjoySpt
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Concept Learning Example (2)

- h , hypothesis, is a set of constraints on attributes:
 - a specific value: e.g., $Water = Warm$
 - any value allowed: e.g., $Water = ?$
 - no value allowed: e.g., $Water = \emptyset$
- Example hypothesis:

<i>Sky</i>	<i>AirTemp</i>	<i>Humidity</i>	<i>Wind</i>	<i>Water</i>	<i>Forecast</i>
$\langle Sunny,$	$?,$	$?,$	$Strong,$	$?,$	$Same \rangle$

Corresponding to boolean function:

$Sunny(Sky) \wedge Strong(Wind) \wedge Same(Forecast)$

- H , hypotheses space, all possible h

Concept Learning Example (3)

- An instance x satisfies a hypothesis h iff all the constraints expressed by h are satisfied by the attribute values in x .

- Example 1:

$x_1: \langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$

$h_1: \langle \text{Sunny, ?, ?, Strong, ?, Same} \rangle$ Satisfies? Yes

- Example 2:

$x_2: \langle \text{Sunny, Warm, Normal, Strong, Warm, Same} \rangle$

$h_2: \langle \text{Sunny, ?, ?, } \emptyset, ?, \text{ Same} \rangle$ Satisfies? No

Least general generalization (1)

- You receive the first instance $\langle \text{Sunny}, ?, ?, \text{Strong}, \text{Warm}, ? \rangle$ and this is your first stage concept
- The second instance is $\langle \text{Sunny}, ?, ?, \text{Strong}, ?, \text{Same} \rangle$ and the concept is changed to $\langle \text{Sunny}, ?, ?, \text{Strong}, ?, ? \rangle$
- The remaining instances do not alter the concept description
- You decide to conclude that Mark enjoys of water sport when is sunny and windy

Least general generalization (2)

- Computing the least general generalization (LGG) of conjunctions due to Plotkin (1970):

INPUT: data D

OUTPUT: logical expression H

$x \leftarrow$ first instance from D

$H \leftarrow x$

while instances left **do**

$x \leftarrow$ next instance from D

$H \leftarrow \text{LGG}(H, x)$ // return literals common to both H and x

end

return H

- Read about the basic extension of computing the LGG of conjunctions employing internal disjunction in Chapter 4 of the main literature (see Algorithm 4.3)

References

- Bruner, J. S., Goodnow, J. J., Austin, G. A. (1956). *A Study of Thinking*. John Wiley & Sons.
- Flach, P. (2012). *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*. Cambridge University Press.
- Plotkin, G. D. (1970). A Note on Inductive Generalization. *Machine Intelligence*. **5**:153-163.